

TASK - 4

Why Visualize Data?

Data visualization means presenting data using charts and graphs so it becomes easier to understand.

Instead of reading large tables of numbers, charts help us:

- Identify trends and patterns
- Compare values easily
- Understand business performance
- Make better decisions

Examples:

- Sales growth over months
- Most sold products
- Regional performance comparison

Visualization makes data simple, attractive, and easier to analyze.

Matplotlib

Matplotlib is a popular Python library used to create charts and graphs.

It helps create:

- Bar charts
- Line charts
- Pie charts
- Histograms
- Scatter plots

Before using Matplotlib:

```
import matplotlib.pyplot as plt
```

Install using:

```
pip install matplotlib
```

Bar Charts

Bar charts are used to compare different categories.

Examples:

- Product sales
- City population
- Student marks

Example:

```
import matplotlib.pyplot as plt

products = ["Laptop", "Phone", "Tablet"]
sales = [50000, 70000, 30000]

plt.bar(products, sales)

plt.title("Product Sales")
plt.xlabel("Products")
plt.ylabel("Sales")

plt.show()
```

Bar charts help compare values side by side.

Line Charts

Line charts are used to show trends over time.

Examples:

- Monthly sales
- Temperature changes
- Website traffic growth

Example:

```
import matplotlib.pyplot as plt

months = ["Jan", "Feb", "Mar", "Apr"]
sales = [10000, 15000, 20000, 25000]

plt.plot(months, sales)

plt.title("Monthly Sales Trend")
```

```
plt.xlabel("Months")  
plt.ylabel("Sales")
```

```
plt.show()
```

Line charts help understand growth or decline.

Pie Charts

Pie charts show parts of a whole using percentages.

Examples:

- Market share
- Expense distribution
- Product contribution

Example:

```
import matplotlib.pyplot as plt
```

```
products = ["Laptop", "Phone", "Tablet"]  
sales = [50, 30, 20]
```

```
plt.pie(sales, labels=products, autopct="%1.1f%%")
```

```
plt.title("Sales Distribution")
```

```
plt.show()
```

Pie charts help understand proportion and contribution.

Complete Data Analysis Workflow

A basic data analysis workflow includes:

1. Collect Data

Data is gathered from CSV files, Excel sheets, databases, or APIs.

Example:

```
import pandas as pd
```

```
df = pd.read_csv("sales_data.csv")
```

2. Explore Data

Understand:

- Rows and columns
- Missing values
- Data types

Example:

```
print(df.head())  
print(df.info())
```

3. Clean Data

Fix:

- Missing values
- Duplicate rows
- Incorrect data

Example:

```
df = df.drop_duplicates()  
df = df.fillna(0)
```

4. Analyze Data

Calculate:

- Total sales
- Average sales
- Best-selling products

Example:

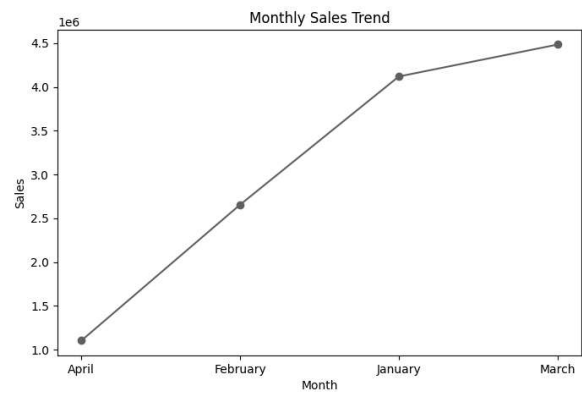
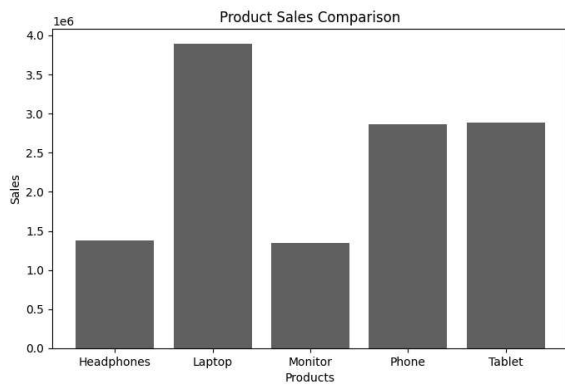
```
total_sales = df["Total_Sales"].sum()  
  
print(total_sales)
```

5. Visualize Data

Create charts and graphs using Matplotlib.

Example:

```
plt.bar(products, sales)  
plt.show()
```



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\91707\Desktop\personal intro project\Week 4> python -u "c:\Users\91707\Desktop\personal intro project\Week 4\main.py"
First 5 Rows:
  Date      Product  Quantity  Price  Customer_ID  Region  Total_Sales
0 01-01-2024 Phone      7    37300    CUST001     East    261100
1 02-01-2024 Headphones 4    15406    CUST002     North    61624
2 03-01-2024 Phone      2    21746    CUST003     West     43492
3 04-01-2024 Headphones 1    30895    CUST004     East     30895
4 05-01-2024 Laptop     8    39835    CUST005     North    318680

===== SALES REPORT =====
Total Revenue      : ₹12,365,048.00
Best Selling Product: Laptop
Charts created successfully!

PS C:\Users\91707\Desktop\personal intro project\Week 4>
```